

# Ricardo Castro Figueiredo

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## Experience

### Critical Manufacturing

Maia, Portugal

Software Developer - Intern

July 2024 – Sept. 2024

- Designed and deployed Feedback Circle, a full-stack web application that streamlined the company's performance appraisal process
- Implemented a component-based architecture using vanilla JavaScript, HTML, and CSS for the frontend, with Node.js and MySQL for the backend
- Designed and built a RESTful API handling user authentication, feedback management, and role-based visibility control
- Created a database schema to manage feedback workflows between employees, managers, and appraisers

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## Projects

- **Kotlin Compiler** *Technologies:* Haskell, Alex/Happy, MIPS
  - Developed compiler with lexer/parser (Alex/Happy), semantic checks, and MIPS codegen for Kotlin constructs
  - Optimized via graph coloring register allocation and liveness analysis
- **CART Class Imbalance** *Technologies:* Python, scikit-learn, SMOTE
  - Boosted F1-score 15% with weighted Gini impurity for minority classes
  - Validated via stratified sampling and ROC-AUC metrics across several datasets
- **Roomba Battle Bot** *Technologies:* Java, Robocode
  - Predictive targeting using trigonometry and adaptive bullet power
  - Energy-efficient evasion via collision timeout tracking + zigzag movement
- **Shinsu Duel: a Tower of God CCG** *Technologies:* Node.js, Socket.IO, Express
  - Real-time PvP/PvE card game with synced state management
  - Component-based UI with drag-and-drop interactions
- **Cryptography Benchmark** *Technologies:* Python, Cryptography, Matplotlib
  - Automated performance analysis of AES/RSA/SHA-256
  - Visualized encryption-decryption trends with statistical averaging

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## Education

### Bachelor in Computer Science

University of Porto

Feb. 2026

- **Coursework:** Algorithms & Data Structures, AI/ML, Computer Systems, Databases & Big Data, Security & Cryptography, Software Architecture, Mathematics, Programming Paradigms, Theory of Computation, Human-Computer Interaction

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## Skills

- **Languages:** Python, Java, C++, Haskell, SQL, JavaScript, HTML/CSS, Bash/Shell, LaTeX
- **Tools & Frameworks:** Pandas, NumPy, MySQL, MongoDB, Node.js, Express.js, REST APIs, TCP/IP, Socket Programming, AMPL, Jupyter, Figma, Git, React
- **Platforms:** AWS, GCP, Linux/Unix, Docker, Anaconda

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## Certifications

- English C1 Advanced (CAE), Cambridge Assessment English

June 2021